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Augmented Experiences in Cultural Spaces through Social Participation

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Abstract

Technologies such as the Internet of Things (IoT) and Social Computing offer promising opportunities to envision and shape augmented experiences in a wide variety of cultural spaces. Such interactive technologies act as a hook to attract visitor's attention, but if they do not provide opportunities to engage people in a more personal and meaningful way, they can be soon discarded. Augmented experiences need to be motivating and connect with values and expectations of visitors and institutions. In a world that is becoming more and more connected and moving towards a participatory culture, the need to participate



<https://dl.acm.org/doi/10.1145/3230675>

The method employed by the Social Display Environment (SDE) is a novel approach to museum engagement that **augments physical exhibition objects with user-generated digital content** to foster social interaction and co-production of meaning.

Here's a short summary of the method and its relevance:

Method: The SDE transforms traditional museum showcases by integrating a **see-through window screen** that superimposes digital information onto physical artefacts. These objects are fitted with **RFID tags** for recognition, allowing the system to display associated content. The core of the method is enabling **user-generated content (UGC)**: visitors can **record short video stories and add text comments** about the objects, expressing their personal interpretations, feelings, or critiques. This content can be created directly at the showcase via a webcam, or, in later iterations, more privately using **personal mobile devices by scanning a QR code**. All contributions are stored and displayed, allowing visitors to **watch, rate, and comment on other visitors' stories**, thereby creating "different threads of discussion" and "new narratives" around the objects. This iterative design process, including formative evaluations with real users, was crucial to refine the system.

Relevancy in the Domain: This method is highly relevant to the cultural heritage domain for several reasons:

- **Promoting Active Participation:** It addresses the growing need for **participatory culture** in museums, moving beyond a unidirectional delivery of knowledge to a model where visitors actively **co-produce knowledge and meaning**.
- **Enhancing Engagement and Connection:** By allowing visitors to express and share their unique views, the SDE taps into intrinsic motivators like the need for **socialization, self-actualization, and social recognition**, fostering a deeper, more personal connection with the exhibits and other visitors.
- **Transforming Objects into "Social Objects":** Inspired by Nina Simon's concept, the SDE turns cultural items into focal points for conversation, enabling them to be **personal, active, or provocative objects** that trigger discussion and shared experiences.
- **Maintaining Physical Connection:** A key design principle is to **avoid "physical disconnection"**; the technology augments the real object rather than creating a separate virtual world, keeping the visitor's attention on the actual cultural item.
- **Enriching Exhibition Spaces:** The user-generated content provides "multiple points of view" and "extraordinarily enriches the exhibition spaces" by allowing professional and unprofessional audiences to "tell stories about cultural heritage".
- **Attracting and Retaining Visitors:** The novelty of the transparent screen acts as a "hook" to attract attention, and the opportunities for involvement, including sensorial, intellectual, and emotional engagement, can transform short-term curiosity into more enduring personal interest.

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MagicARTS: an interactive social journey in the art world

Authors: Kyriakos Christodoulou, Maria Vayanou, George Tsampounaris, Yannis Ioannidis | [Authors Info & Claims](#)

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Abstract



Considering the virtual museum visit as a group activity, in this paper we explore how to support and facilitate social interactions between visitors in the 3D environment of an interactive virtual museum, while also accounting for isolated moments of personal



<https://dl.acm.org/doi/10.1145/3291533.3291579>

MagicARTS is a **multi-user virtual gallery** designed to **facilitate social interactions** among visitors in a 3D environment, recognising that people typically visit cultural heritage sites in groups. It moves beyond traditional virtual museums by not just offering narratives and multimedia, but also allowing visitors to **"dive into the virtual world of the artworks"** and see depicted characters come to life.

Key aspects of MagicARTS include:

1. **Social Experience:** Visitors can **talk with each other**, **share their feelings** about artworks, point to exhibits, and invite others nearby in a playful way. It supports **voice-communication**, **gestures** by avatars, and **audio-based cues** signalling other visitors' actions.
2. **Immersive Content:** It blends 2D artwork representations with **audios**, **animations**, and **immersive 3D models** of paintings to offer a **multi-sensory experience**.
3. **Interactive Artworks:**
 - **Animated Paintings:** 2D or 3D animated paintings with motion and/or sound effects are **activated when a visitor approaches** them (indicated by red semicircles on the floor). These animations are **projected on shared areas**, visible to all group members, and accompanying audio is synchronously delivered to the group, fostering a **social viewing experience**.
 - **3D Environments:** Paintings can act as "portals," transferring visitors into **3D models of the artwork's environment**, allowing them to jointly explore the scene, like Vincent Van Gogh's "The Bedroom".

4. **Emotion Sharing:** MagicARTS is innovative in prompting visitors to **identify and share their personal feelings** about artworks using "emotion bubbles" (emoticons and thought bubbles) that appear over their avatars and are **visible to the whole group**, creating a point of discussion.

5. **Personal Engagement & Signalling:** While promoting social interaction, it also accounts for **private moments** by delivering selected audio/video content exclusively to the initiating visitor. This is non-intrusively signalled to others by the visitor's avatar visually "wearing headphones and a head mounted display".

The system aims to enhance co-presence and interaction by offering a **shared virtual space** that is collaboratively explored, with digital resources gradually unlocked and content remaining on display after interaction, indicating prior engagement.

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Exhibiting emotion: using digital technologies to discover emotional connections

Author:  Genevieve Alelis | [Authors Info & Claims](#)

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Abstract

 This paper discusses the ongoing development of an engaging system that will allow museum visitors to understand their emotional connections to artefacts. Through

<https://dl.acm.org/doi/pdf/10.1145/2468356.2468538>

This paper, "Exhibiting Emotion: Using Digital Technologies to Discover Emotional Connections," is highly relevant to your domain of "Enhancing Gallery Experience through SoMo, building connection," particularly in its emphasis on foregrounding emotions.

- A key finding of the study is that **when prompted to provide emotional responses to artefacts, visitors are motivated to understand the reasons behind these responses, leading to meaningful and personal connections**. This directly reinforces the value of centering emotions in your domain, as it indicates a natural human desire for deeper engagement with artwork beyond mere observation.
- The paper aims to address a gap in previous studies by focusing on the **personal reasons for an emotional response** to artefacts. It acknowledges that while emotions influence daily decision-making, people often do not identify the cause of their emotional responses. To fill this gap, the study conducted interviews to understand the types of emotional connections museum visitors make. The findings suggest that these emotional connections **transcend age and gender**, making the **sharing and understanding of similar and contrasting connections highly valuable** for translation into a mobile application.
- For system design, the paper advocates for a **manual-entry approach to emotion input** rather than automatic detection methods (like biometric sensors), which were found to be invasive and expensive. Manual input encourages users to **reflect on their responses, recall related memories, or connect the artwork to something personal in their lives**, leading to a deeper understanding of their emotional connection. This aligns with the domain's intent to translate feelings into "shareable, in-situ media" and foster "chain-based co-created music," as it relies on active user participation and reflection.
- Ultimately, the paper envisions an ideal system that is **engaging, allows visitors to become aware of their emotions, accepts user-generated feedback**, and facilitates

social connections. This directly supports your domain's goal of designing social and context-aware interactions that enable people to connect more deeply with art and with each other, by making visitors' emotional interpretations a central, shareable component of the gallery experience.

The **manual method** used in this study to capture visitors' emotions was an **Emotional Response Log**.

Here's how it worked:

- **Format:** The log was an **A5-sized booklet** given to each participant at the start of their museum visit.
- **Instructions:** Participants were asked to provide their **initial emotional responses for up to 5 museum artefacts** that generated a strong emotional reaction.
- **Sections:** The log contained dedicated sections for:
 - **Gallery Number**
 - **Artefact Name or Number**
 - **Emotion Felt**
 - **Additional Comments**
- **Emotion Choices:** Within the "Emotion Felt" section, visitors could choose from **six emotions based on Ekman's research** on universal facial expressions: **Anger, Disgust, Fear, Happy, Sad, and Surprise**.
- **Additional Options:** Two further options were included: **"Indifferent"** for neutral responses, and **"Other"**, allowing visitors to write in any emotions not already listed.
- **Flexibility:** Visitors were permitted to **choose as many emotions as they felt** in response to a single artefact.

This manual input method was chosen to encourage users to reflect on their responses, recall related memories, or connect the artwork to something personal in their lives, leading to a deeper understanding of their emotional connection.
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